

Project Design Phase
Proposed Solution Template

Date	1 October 2026
Team ID	
Project Name	FitBuddy
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Generic workout routines fail to adapt to a user's specific changing needs or physical feedback, causing individuals to lose motivation and abandon their fitness goals due to rigid, unstructured, and unpersonalized regimens.
2.	Idea / Solution description	FitBuddy is a FastAPI-based web application that leverages Google Gemini AI models to generate personalized 7-day workout plans and nutrition tips. It features a continuous feedback loop, allowing users to submit text feedback (e.g., "more cardio") to dynamically update and refine their active routines based on their evolving fitness journey.
3.	Novelty / Uniqueness	Unlike static fitness apps, FitBuddy utilizes a dual-model AI architecture (Gemini Pro for complex, day-by-day plan generation and Gemini Flash for rapid nutrition/recovery tips) combined with an iterative feedback mechanism that mutates the user's specific plan over time rather than starting from scratch.
4.	Social Impact / Customer Satisfaction	Democratizes access to highly personalized fitness coaching. By offering structured, goal-aligned routines and dynamic adjustments, it increases user motivation and adherence, leading to better long-term physical health outcomes without the prohibitive cost of human personal trainers.
5.	Business Model (Revenue Model)	A freemium Software-as-a-Service (SaaS) model. Basic users receive standard 7-day plan generation and initial nutrition tips. A premium tier (subscription-based) unlocks unlimited AI feedback iterations, advanced recovery tracking, and priority prompt processing.
6.	Scalability of the Solution	Highly scalable due to the asynchronous, lightweight FastAPI backend and stateless AI generation via the Google GenAI SDK. While currently utilizing SQLite for local storage, the SQLAlchemy database models can be instantly migrated to PostgreSQL or cloud databases to handle thousands of concurrent users.